



BHABHA UNIVERSITY, BHOPAL(M.P)

PROGRAMME NAME.... MECHANICAL ENGINEERING(DIPLOMA ENGINEERING)

SCHEME OF STUDIES AND EXAMINATION FOR : THIRD SEMESTER

Session:2019

COURSE CODE	COURSE TITLE	THEORY COMPONENT												TOTAL CREDIT	GRAND TOTAL MARKS
		LECTURE	CONTINEOUS EVALUATION		END OF THE TERM/SEMESTER EVALUATION		THEORY CREDIT	PRACTICAL PER HOURS	CONTINEOUS EVALUATION	END OF THE SEMESTER		PRACTICAL CREDIT			
			Hrs PER WEEK	TERM WORK, QUIZ,AS SIGNMENT	MID TERM					THEORY PAPER			LAB WORK, QUIZ,ASSIGNMENT		
		I	II	MARKS	DURATION(Hrs)										
ME-3001	MATERIALS TECHNOLOGY	4	10	10	10	70	3	4	3	20	30	3	2	6	150
ME-3002	MANUFACTURING PROCESS	4	10	10	10	70	3	4	4	20	30	3	2	6	150
ME-3003	BASIC ELECTRICAL& ELECTRONICS	3	10	10	10	70	3	4	2	20	30	3	2	6	150
ME-3004	MECHANICAL DRAFTING & AUTO CAD	2	10	10	10	70	3	4	6					4	100
ME-3005	STRENGTH OF MATERIAL	4	10	10	10	70	3	4	2	20	30	3	2	6	150
TOTAL		17	50	50	50	350		20	19	80	120		8	28	700



DIPLOMA IN MECHANICAL ENGINEERING

ME-3001 MATERIALS TECHNOLOGY

THIRD SEMESTER

S.NO	COURSE CONTENT	STUDY HOURS
1.	Requirement of Engineering materials, mechanical properties and their testing : Introduction to engineering materials, classification of engineering materials and their properties. Mechanical properties of materials, destructive including Tensile test, compression test, hardness test, impact test fatigue test, endurance limit, bending test, shear test and non- destructive testing methods.	03
2	Structure of Solid materials: Classification amorphous and crystalline states, unit cells and crystal structure (B.C.C., F.C.C. and H.C.P) allotropy. Crystal imperfection and their effects on properties	02
3	Solidification of Metal and ingot structure: Process of nucleation and grain growth, ingot solidification, dendritic and columnar structure, segregation of impurities, grain and grain boundaries.	04
4	Equilibrium Phase Diagrams and Phase Transformation : Equilibrium of phase Diagrams : Plotting of equilibrium diagrams, interpretation, phase rule and lever rule and its application Phase transformations – Eutectic Eutectoid, Peritectic and Peritectoid	06
5	Practical Metallography : Preparation of specimen, selecting the specimen, mounting the specimen, grinding , polishing, etching and etching reagents. The metallurgical microscope. Use and care of microscope.	02
6	Iron- Carbon Equilibrium System : The complete iron carbon diagram and its interpretation. The solidification and cooling of various carbon steels, structures produced, correlation of mechanical properties with carbon content.	06
7	Heat Treatment of Steels: Objective of heat treatment, thermal processes- annealing, normalizing, hardening and tempering. Hardening process : Surface hardening, flame hardening, case	05



	hardening methods, their scope, limitations and advantages, quenching mediums and their effect on hardness, Hardening defects due to improper quenching, hardenability, Jominy end quench test and interpretation of its results. T.T.T. curves interpretation and use, Isothermal heat treatment processes -martempering, austempering, spheroidising and patenting	
8	Ferrous Metals and Alloys : Classification, types of cast irons their properties and uses, alloy cast-irons, various alloying elements used, their effects on properties and uses. Classification, composition and uses of plain carbon steels, effect of impurities, Alloy steels -various alloying elements, their effects on properties and uses. Alloy steel classification.Tool Steel : Typical compositions, requirements of tool steels, high speed steel, high carbon steel. Standardization of steels. Designation of steels as per B.I.S. codes.	05
9	Non- Ferrous Metals and Alloys : Copper : Its Properties and uses Cooper Bases Alloys : Brasses, their classification, composition, properties and uses, designation of copper alloys as per B.I.S. aluminum its properties and uses. Aluminum Alloys: Their composition, Classification, properties and uses. Designation of Al- alloys as per B.I.S, Zinc, Nickel and lead their alloys properties and uses Bearing alloys - their composition and field of application.	05
10	Non- Metallic Materials : introduction to Ceramic Refractory, Rubbers Insulators and Lubricants	03
11	Plastics : characteristics, classification, commonly used thermo-setting and thermoplastic - their properties and uses. Ingredients for processing plastics. Plastic processing methods different methods.	06
12	Powder Metallurgy :-Introduction and application. Description of process, manufacture and blending of metal powder compacting and sintering.	03
13	Metal Preservation : Corrosion meaning various mechanism effect of corrosion, methods of minimizing corrosion	06
14	Modern Trends in Materials Engineering : New materials like FRP, Composites, synthetic fibers, synthetic wood. Super conductors	04



DIPLOMA IN MECHANICAL ENGINEERING

ME-3001 MATERIALS TECHNOLOGY

THIRD SEMESTER

LIST OF EXPERIMENTS

1	Preparation of micro specimen.	03
2	To study micro structural characteristics of gray cast iron white cast iron and malleable cast iron.	06
3	To study effect of normalizing, annealing on the hardness and microstructure of high carbon steel.	06
4	To study the effect of carbon and temperature on hardening of steel.	06
5	To study the effect of temperature on the properties during tempering of steel.	06
6	To study the effect of quenching media on hardness of steel.	03
7	To study the carbonizing and case hardening of steel.	03
8	Joining hardenability test and its industrial use.	03
9	To Study the microstructure of some important brasses and bronzes.	03
10	To observe the micro structural characteristics and other properties of various cast irons and prepare a report for industrial uses.	06



DIPLOMA IN MECHANICAL ENGINEERING

ME-3001 MATERIALS TECHNOLOGY

THIRD SEMESTER

REFERENCES

1. Engineering physical Metallurgy-By Prof. Y Lakhtin MIR Publishers Moscow
2. A Text Book of Material Science And Metallurgy by O.P. Khanna.
3. Material Science And Process. by S. K. Hazia Choudhry
4. Mechanical Metallurgy by Dieter (Tata Mcgrawhill)
5. Materials For Engineers by M.H.A. Kempsty
6. Introduction to Material Science And Engineering by K.M. Ralls, T.H. Courtney, John Wuff (Wiley Eastern New Delhi)
7. Physical Metallurgy Principles by Read Hill (Affiliated East- West Press Pvt. Ltd. New Delhi.)
8. Engineering Metallurgy by R. Higgins (ENS).
9. Materials Science by B.S. Narang (Pub. CBS pub. & Distributions New Delhi)
10. Padarth Prodyogiki (Hindi) by P.N. Vijayvergiya (Deepak Prakashan, Gwalior) .



DIPLOMA IN MECHANICAL ENGINEERING

ME-3002 MANUFACTURING PROCESS

THIRD SEMESTER

S.NO	COURSE CONTENT	STUDY HOURS
1	Introduction to Manufacturing Processes : Definition, classification of basic manufacturing processes i.e, mechanical working, casting, metal joining processes, metal cutting process, press working . Examples of each of the above listed manufacturing processes, factors which influence selection of manufacturing process for a particular application.	03
2	Metal Casting : Introduction , advantages and limitations of casting as production process. Pattern Making : Definition of pattern, types of patterns and their details, materials, allowances, tools required, colour code for patterns. Moulding : Definition, moulding methods and types of moulds, moulding materials, moulding sand and its composition, sand properties, testing parameters of sand, and their effects, sand preparations, sand conditioning, characteristics and defects of moulds. Function of runners, risers and gate. Cores and core making, core boxes. Cleaning of casting, Special casting methods, need for special casting methods, die casting, centrifugal casting, investment (lost wax) casting, casting defects, causes and analysis, area of application of casting process. Furnaces: Cupola ,crucible, pit and electric arc furnaces, induction furnace , their salient features, safety aspects.	18
3	Press Working : Introduction of press working of metals, principle of press working , description of a simple press working unit, press working operations punching, shearing, drawing, bending, slitting, knurling, notching, trimming, piercing etc. Double action press, description and its field of application, die and punch, types of dies, specifications of a press, safety precautions to be observed while working on a press.	15
4	Mechanical Working : Introduction - hot and cold working	10



	Principle of re-crystallization. Metal Rolling : Principle of metal rolling , basic components of a simple rolling process equipment. Types of deformation during rolling. roller material, selection and desirable properties, principles of thread rolling-description with sketches, manufacture of seamless tubes by rolling. types of rolling mill. Rolling defects Metal Drawing : Basic Principle of drawing of metals, differentiate between the drawing and deep drawing of metals, principle of wire drawing and example. Extrusion : Definition, Classify the methods of extrusion, their limitations, advantage and disadvantage. Tube extrusion, impact extrusion, application of extrusion processes. Extrusion defects . Forging : Types of forging, Die forging, differentiate between the cold die and hot die forging , advantage of forming by forging, common defects and their reasons. Limitations of forging, press forging, drop forging, upset forging, die material , applications of forging processes in engineering.	
5	Metal Joining : Introduction, Classification of metal joining processes Welding :-classification, Plastic, fusion and forge welding, Weldability of metals, metallurgy of welding Resistance welding : Spot, seam, butt, projection, percussion techniques. Gas welding and gas cutting : Principle of operation and technique, gas cutting. Arc Welding : Carbon arc, TIG, MIG, Submerged arc , Atomic hydrogen, Eletro-slag, Plasma arc welding processes, Electrodes- types and selection , flux and their uses. Special welding techniques-Welding of different metals. Defects in welds, testing and inspection. Accident prevention in gas and arc welding Equipments & tools used in metal arc welding, specification and functions. Soldering, Brazing and Adhesive bonding	14



DIPLOMA IN MECHANICAL ENGINEERING

ME-3002 MANUFACTURING PROCESS

THIRD SEMESTER

LIST OF EXPERIMENTS

S.No.	NAME OF EXPERIMENT	SHOP	PRACT. Hrs.
1	Making a split/solid pattern from wood. Making a core box.	Carpentry /pattern shop	06
2	Tempering of sand, practice of green and dry sand making.	pattern shop	03
3	Practice of core making and baking	Moulding shop	03
4	Practice of open mould in a two boxes, using split pattern and solid pattern, Locating the core .	Moulding shop	03
5	Demonstration of metal melting in pit furnace& casting process.	foundary	03
6	Simple forming practice (Making a square bar out of a given round bar, making of a chisel and bolt)	Blacksmith shop	03
7	Practice of upsetting of a round on power hammer.	Blacksmith shop	03
8	Practice of sheet cutting with the help of straight and bent snips. Making small rectangular prism and cylinder.	Blacksmith shop Tinsmith	09
9	Practice of making of washer of any size on a flypress.	Blacksmith shop Tinsmith	03
10	Practice of piercing, notching and circle cutting with the help of Metal master machine.	Blacksmith shop Tinsmith	06
11	Practice of piercing, notching and circle cutting with the help of Metal master machine.	Blacksmith shop Tinsmith	03
12	Practice of piercing, notching and circle cutting with the help of Metal master machine.	Blacksmith shop Tinsmith	03
13	Practice of edge preparation for welding.	Fitting	09
14	Demonstration and practice of bead laying (Welding) on a Flat pieces	Fitting	03



15	Practice of Welding of corner, edge and Tee joint	Welding	02
16	Welding ' V' butt joint.	Welding	03
17	Practice of joining wires and rods of different size on spot welding machine.	Welding	03
18	Practice of making gas flames with nozzles and making simple joints.	Welding	02



DIPLOMA IN MECHANICAL ENGINEERING

ME-3002 MANUFACTURING PROCESS

THIRD SEMESTER

REFERENCES

1. Process And Materials of Manufacture by Lindberg.
2. Workshop Technology by Hazara & Choudhary.
3. Materials And Manufacturing process by Dalela.
4. Manufacturing Processes by Yankee.
5. Manufacturing Process by S.E. Rusinof
6. Welding Engineering by B.E. Rossi.
7. Production Engineering – P.C. Sharma
8. Manufacturing Technology- P.N. Rao
9. Production Technology- R.K. Jain
10. Foundry Engineering by P.L. Jain.
11. Nirman Prakram (Hindi) by P.N. Vijayvargiya. (Deepak Prakashan, Morar, Gwalior)



DIPLOMA IN MECHANICAL ENGINEERING

ME-3003 BASIC ELECTRICAL & ELECTRONICS

THIRD SEMESTER

S.NO	COURSE CONTENT	STUDY HOURS
1.	FUNDAMENTALS OF ELECTRICAL ENGINEERING Concept of electric current, potential and potential difference (Voltage). Sources of D.C. and A.C. Electric energy. Methods of voltage generation and standard voltages used in generation transmission and distribution. Electrical Power, energy and their units.	2
2	D.C. CIRCUITS Ohm's Law, Concept of resistance, conductance, resistivity, conductivity and their units. Effect of temp. on resistance. Temperature coefficient of resistance (Definition only) Connections of resistances. Series, Parallel connections and their combinations. (Simple Numericals) Kirchoff's Voltage Law, Kirchoff's Current Law (Simple Numerical)	4
3	A.C. CIRCUITS Generation of single phase and three phase sinusoidal voltage. Vector representation. Concept of Cycle, Frequency, time period, amplitude, phase and phase difference. Define instantaneous value, average value, RMS value and peak value of sinusoidal electrical quantities. Derive relationship between them . Form factor and peak factor (Definition only). Current voltage and power in pure resistive, inductive and capacitive circuit. Concept of Reactance , impedance and power factor in R.L., R. C. and RLC Series circuit. (Simple Numericals). Causes and effect of poor power factor. Methods of improving power factor. phase AC supply- three phase three wire and three phase four wire system. Relationship between VL and VPH, IL and IPH and threephase power in star and delta connected load.(Simple Numerical)	5



4	D.C. MACHINES Review of concept of Electromagnetism and related laws (Faraday's Law, Lenz's Law, Cork Screw Rule, Fleming's Left Hand & Right Hand Rule.) Construction of D.C. Machines, its main parts and their functions. Classification of D.C. Machines. D.C. Generator : Principle , EMF equation(Derivation and simple numerical), Types of D.C. Generator (No Numerical), Applications of DC generator. D.C. Motor : Working Principle of DC motor, Types of DC motor, significance of back emf, Torque equation(No Numerical). Characteristics of D.C. Series and Shunt Motor. Applications of D.C. Motor. D.C. Motor starter - 3 point starter. Efficiency (No numerical)	8
5	A.C. MACHINES Single Phase Transformer : Construction, working principle. EMF Equation (Derivation with simple numerical), Turn ratio, Step up and step down transformers and their application. Losses , efficiency and regulation (No derivation). Three Phase Induction Motor : Construction, types , principle of operation. Concept of Slip (Simple Numerical), Applications, Starters: DOL and Star Delta. Single Phase Induction Motor : Methods of making a single phase motor self start. Types of single phase induction motor- capacitor start, capacitor run and shaded pole and their applications. Synchronous Machines : Synchronous motor- construction, principle of operation, comparison with three phase induction motor. Synchronous generator (alternator) – Construction, principle of operation, speed and frequency, synchronous speed.	8
6	MEASURING INSTRUMENTS AND TRANSDUCERS Classification of Measuring Instruments, absolute and secondary instruments. Indicating, Integrating and Recording instruments, their examples. Elementary idea about working principles and construction of MI and MC type Ammeter and voltmeter. Electrodynamicometer type watt meter. Induction type energy meter , electronic energy meter. Application of Megger and earth tester. Multimeter, CRO, its block diagram and applications. Transducers – Definition, primary and secondary transducers, active and passive transducers, variable parameter R,L,C type transducers. General idea about strain gauge, LVDT, Thermocouple, Piezo Electric and Photoelectric Transducers.	6



7	ELECTRIC WIRING Types of Wiring and their Applications. Size of conductor, S.W. gauge. Accessories like switches, fuses, holders, sockets and MCB's. Staircase Wiring, Fluorescent tube light wiring.	2
8	ELECTRONIC DEVICES AND CIRCUITS Semiconductor PN Junction Diode, Zener Diode, PNP and NPN transistor, UJT, FET, MOSFET and SCR. Their layer diagram, symbol, V-I characteristics and applications. Electronic Circuits: Concept of biasing of diode and transistor. Single Phase Half wave and Full wave rectifier (I/O waveform), Concept of ripple, filter circuit (shunt capacitor and series inductor). Transistor as an amplifier, concept of gain, Zener regulator, regulated power supply (Block diagram only).	8
9	ELECTRIC SAFETY 9.1 Electric shock and its prevention , effect of electrical current on human body, shock treatment, need of earthing.	2



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ME-3003 BASIC ELECTRICAL & ELECTRONICS

THIRD SEMESTER

LIST OF EXPERIMENTS

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2	To study micro structural characteristics of gray cast iron white cast iron and malleable cast iron.	06
3	To study effect of normalizing, annealing on the hardness and microstructure of high carbon steel.	06
4	To study the effect of carbon and temperature on hardening of steel.	06
5	To study the effect of temperature on the properties during tempering of steel.	06
6	To study the effect of quenching media on hardness of steel.	03
7	To study the carbonizing and case hardening of steel.	03
8	Joining hardenability test and its industrial use.	03
9	To Study the microstructure of some important brasses and bronzes.	03
10	To observe the micro structural characteristics and other properties of various cast irons and prepare a report for industrial uses.	06
S.No.	NAME OF EXPERIMENT	STUDY HOUR S
01.	Verification of Kirchoff's Law	
02.	To Measure Voltage Current and power in single phase AC circuit.	
03.	To calculate Impedance, power and power factor by measuring	



04.	voltage across each element and current to the circuit for a given	
05.	RLC series circuit.	
06.	Study of different parts of DC machine.	
07.	Study of three point starter for DC motor.	
08.	To determine transformation ratio of a given single phase	
09.	Transformer.	
10.	Study and operation of DOL and Star Delta Starter.	
11.	To measure slip for a given three phase induction motor.	
12.	To measure insulation resistance by Megger.	
13.	To measure earth resistance by earth tester.	
14.	To measure linear displacement by LVDT.	
15.	To make connection for stair case wiring.	
16.	To make connection for fluorescent tube light circuit.	
17.	Testing of various electronic components using millimeter and CRO.	
18.	To draw V-I characteristics of Zener diode.	



19.	To observe input and output wave form of full wave / half wave rectifier on CRO.	
20.	To observe amplified output of a given amplifier.	



DIPLOMA IN MECHANICAL ENGINEERING

ME-3003 BASIC ELECTRICAL & ELECTRONICS

THIRD SEMESTER

REFERENCES

1. Fundamental of Electrical Engineering and Electronics – B.L. Thareja , S. CHAND Publication
2. Basic Electrical Engineering – V.K. Mehta , S. CHAND Publication
3. Principles of Electronics – V.K. Mehta , S. CHAND Publication
4. Basic Electrical Engineering – V. N. Mittle, TMH
5. Electrical Machines Vol I & II – S.K.Bhattacharya, TTTI, Chandigarh
6. A Curse book in Electrical & Electronic Measurement and Instrumentation , A.K.Sawhney.
7. Principles of Electrical Engg. and Electronics – V.K. Mehta , S. CHAND Publication
8. Electrical Technology – S.L.Uppal, Khanna Publication
9. EElectrical Measurement – J.B. Gupta, Dhanpat Rai & Sons.



DIPLOMA IN MECHANICAL ENGINEERING

ME-3004 MECHANICAL DRAFTING AND AUTOCAD

THIRD SEMESTER

S.NO	COURSE CONTENT	STUDY HOURS
1	Projection and multi view Representation: Projection orthographic projection. First and third angle projection, superfluous view, choice of views, auxillary views- views -full and partial, conversion of pictorial views in to orthographic views, conventional representation as per IS:696.	1
2	Sectional Views : Full section, half section, partial or broken section, revolved section, removed section, offset section. Sectioning conventions, section lines. Hatching procedure for different materials as per IS code 686 1972. Sectional views of assembled parts. Choosing from IC engine parts, couplings, clutches, brackets, bearing etc. (Use 1st and 3rd angle projections both)	1
3	Dimensioning Tolerance, Machining And Welding Symbols : Types of dimensions (size and location) dimensioning terms and notations. (use of I.S.Code 696 & 2709) general rules for dimensioning and practical hints on dimensioning systems of dimensioning. Dimension of cylinder holes arcs of circle narrow space, angles, counter sunk hole, screw threads taper etc. Application of tolerances. (Use I.S. Code 696) Machining marks, finish marks, countersinking, counter boring spot facing, figures and notes for same. Representation of characteristics machining (circularity, Angularity etc .) (Ref IS 969) Representation of welded joints, welding symbols, tolerance of forms and positions. Procedure of drawing fits, limits, size, tolerance, clearance etc.	1
4	Production Drawing: Detailed drawing, assembly drawing, scale, finish tolerances, notes etc. Title block, tool list, gauge list. Preparation of production drawing for pattern shop. Forging shop, machine shop, preparation of assembly drawing from detailed drawing. exploded views, sectional pictorial views, assembly drawing of nut and bolt, plumber block, flange coupling, stepped pulleys, foot-step bearing, Universal	10



	coupling, connecting rod, piston of I.C. engines, cotter joint, Knuckle joint. Preparation of detailed drawing from assembly drawings and assembled pictorial views, Interpretation of production drawing.	
5	Introduction to Auto CAD: Coordinate system. Draw command-line ,arc, circle rectangle, polygon, point, ellipse, hatch, table. Modify commands-erase, copy, offset, array, trim, extend, break, join, chamfer, fillet, move, rotate, scale, stretch, lengthen. Dimensioning Tray settings: snap, grid, ortho, polar, osnap Format commands: line type, point style, units, layers, drawing limit, dimension style	06
6	Application of Auto CAD: practice of assembly drawings using Auto CAD	04
7	Presentation: Block, creating layout, insert layout ,plotting/printing	02
8	Pipe Drafting : Various symbols used in pipe line work as per IS code of Practice, C.I. flanged joint, socket and spigot joint, gland and stuffing box, expansion joint, pipe fitting typical pipe bends, pipe supports and accessories.	02
9	Gear Drawing : Gear terminology such as pitch, pitch circle diameter module, addendum, root circle diameter, hole depth, blank diameter etc. construction of cycloidal, involute teeth profiles, pinion and rack meshing, spur gear meshing.	02
10	Graph and Charts : Introduction, Classification of chart, graphs and diagrams, quantitative and qualitative charts and graphs, Drawing and curve titles, legends notes etc. procedure for making a graphical representation in ink. Logarithmic graphs, semi logarithmic graphs, bar charts area (Percentage) charts, pie chart, alignment charts (Nomo graphs) Forms and construction, construction of functional scale, parallel scale charts for equations of the form $f(t) + f(u) + f(v)$, $(f(t) f(u) = f(v))$ three scale alignment chart, graphical construction of a Z- chart, four variable relationship parallel scale alignment chart.	01



DIPLOMA IN MECHANICAL ENGINEERING

ME-3004 MECHANICAL DRAFTING AND AUTOCAD

THIRD SEMESTER

SUGGESTED TERMS –WORK

S.No.	TOPIC	PRACT. SHEETS IN NOS.	HOURS OF STUDY
1	Projection and multi views representation	02	1
2	Sectional views	02	2
3	Dimensioning, tolerance, machining and welding symbols	02	3
4	Production drawing	08	4
5	Pipe drafting	01	5
6	Gear Drawing	02	6
7	Graphs and charts	01	7
8	Computer graphics	05	8
	TOTAL	23	90



DIPLOMA IN MECHANICAL ENGINEERING

ME-3004 MECHANICAL DRAFTING AND AUTOCAD

THIRD SEMESTER

REFERENCES

- 1 Fundamentals of Engineering Drawing by Warren J. Luzadder (Prentice-Hall).
- 2 Mechanical Drawing by Giesecke, Mic- hell Specer, Hill. (Collier Macmillan Internal Edition)
- 3 Engineering Graphics by Giesecke/Mitchell/ Spencer/ Hill/ Loving (Macmillan) .
- 4 Mechanical Drawing By N.D.Bhatt
- 5 Mechanical Drawing By P.S.Gill
- 6 Mechanical Drawing By R.K.Dhawan
- 7 Inside AUTO CAD by Daniel Raker and Harbest Rice(BPB Publisher)
- 8 Computer Graphics and CAD Fundamentals By Noel M Morries(Wheeler)



DIPLOMA IN MECHANICAL ENGINEERING

ME-3005 STRENGTH OF MATERIAL

THIRD SEMESTER

COURSE CONTENTS

S.NO	COURSE CONTENTS	STUDY Hrs.
1	Simple Stress and Strains : Introduction types of loads and deformation, types of stresses and strain. Hooke's law, stress strain diagram for ferrous and non ferrous materials modulus of elasticity. rigidity and bulk modules of materials Stress in bars of varying cross sections, composite sections and compound sections Thermal stresses and strains, thermal stresses in composite sections. Poisson's ratio, volumetric strain, relation between different modulus, strain energy, resilience, proof resilience, modules of resilience suddenly applied loads and impact loads.	06
2	Mechanical properties and their testing : Mechanical properties of materials, destructive including Tensile test, compression test, hardness test, torsion test, impact test fatigue test, endurance limit, bending test, shear test and non- destructive testing methods.	04
3	S.F. and B.M. Diagrams : Definition, types of loading types of beams, shear force and bending moment sign conventions S.F. and B.M. diagrams for cantilever simply supported and overhanging beams with point or concentrated loads uniformly distributed loads and combination of point and U.D.L. Point of contra flexure, numerical problems.	06
4	Principal Planes and Principal Stresses : Stresses on inclined plane subjected to direct shear or combination of stresses in two mutually perpendicular planes. Principal planes and principal stresses, analytical and graphical methods.	06
5	Bending Stresses in Beams : Theory of simple bending as assumptions made in simple bending theory position of neutral axis, surface moment or resistance. Modules of section of	06



	symmetrical sections such as rectangular, circular and I sections, bending stresses in symmetrical sections. Simple problems. Reinforced concrete beams, beam of uniform strength.	
6	Shear Stresses in Beams.: Introduction shear stress equation, assumptions made, distribution of shear stresses over various sections, such as rectangular, circular and I L & T sections, Simple numerical problems.	04
7	Deflection of Beams : Introduction Strength and stiffness of beam curvature of bent beam, Derivation of equation for slope and deflection of beam in case of cantilever and simply supported beam loaded with point loads U.D.L. and combination. Simple numerical problems. Importance of deflection and practical applications.	06
8	Torsion of Shaft : Definition of torsion relation between stress, strain and angle of twist assumptions made strength of solid and hollow circular shaft, polar moment of inertia. Calculation of shaft diameter on the basis of strength and stiffness for the given horse power transmitted torsional rigidity. Maximum torque comparison of solid and hollow shaft size of a shaft for a given torque.	04
9	Spring : Definition types and use of springs, leaf spring, helical and spiral springs, Stiffness of a spring and maximum shear stress, defection of spring . Spring Classification based on size shape and load.	04
10	Columns and struts : Definitions crippling load different end conditions, slenderness ratio, equivalent length, Euler's theory Rankine's formulae, radius of gyration, Rankine constant for different materials Limitations of Rankine formula simple problem B.I.S. code for columns.	04
11	Stresses in Frames : Definition of frame, perfect, deficient and redundant frame. Assumptions made in finding stress in method of sections and graphical method Bows notation, solution of problems using three methods.	06
12	Thin Cylinders and Spheres : Hoop stress longitudinal stress on inclined plane subject subjected to direct, shell, volume strain change in value, cylindrical vessels subjected to internal pressure, simple numerical problems.	04



DIPLOMA IN MECHANICAL ENGINEERING

ME-3005 STRENGTH OF MATERIAL

THIRD SEMESTER

LIST OF EXPERIMENTS

S.No.	NAME OF EXPERIMENT	PRACT. Hrs
01	Study and demonstration of Universal Testing Machine & its attachments	02
02	Tension Test on mild steel, Aluminium & compression test on cast iron on Universal Testing Machine.	03
03	Direct Shear Test of mild steel on Universal Testing Machine.	03
04	Brinell Hardness Test on Mild Steel.	02
05	Rockwell hardness Test on Hardened Steel.	02
06	Izod & Charpy - Impact tests of a standard specimen	03
07	Torsion Test on Mild steel bar.	03
08	Drawing sheet on shear force & bending Moment diagrams for a given loading (At least four problems.).	06
09	Estimation of principal stresses and maximum shear strain for a given combined loading by analytical & Mohr's circle method. (At least two problems.).	06



DIPLOMA IN MECHANICAL ENGINEERING

ME-3005 STRENGTH OF MATERIAL

THIRD SEMESTER

REFERENCES

- 1 Strength of Materials. by B.C. Punmia.
- 2 Strength of Materials . by R.S. Khurmi.
- 3 Strenght of Materials by Sadhu Singh.
- 4 Strength of Materials by K.D. Sexena.
- 5 Strength of Materials by S. Ramamuruthan.
- 6 Strength of Materials by I.B. Prasad.
- 7 Strength of Materials by Ryder.
- 8 Strength of Materials by Timo shanko & young
- 9 Laboratory Experiments In Strength of Materials by B.D. Sharma.
- 10 Dravya Samarthy (Hindi) by K. D. Saxena (Deepak Prakashan, Morar Gwalior)